MTH 202 Chapter 3: Partial Derivatives

3.9 Lagrange Multipliers Worksheet

1. Suppose ACME's revenue (in thousands of dollars) from the sale of x widgets and y gadgets can be expressed by $R(x,y) = 150x - x^2 + 120y - \frac{1}{2}y^2$. Suppose that creating a widget requires 4 gizmos and gadget requires 2 gizmos. ACME only has 120 gizmos available and wants to use all of them. Determine ACME's maximum revenue subject to the given constraint.

2. Find the points on the surface $xy - z^2 = 1$ that are closest to the origin.

3. Let $T(x,y,z)=20+2x+2y+z^2$ represent the temperature at each point on the sphere $x^2+y^2+z^2=11$. Use Lagrange multipliers to find the extreme temperatures on this surface.

4. Use Lagrange multipliers to find the maximum and minimum values of the function $f(x, y) = x^2 + y^2 + z^2$ subject to the contain $x^2 + y^2 + 2z = 8$.

5. Find the points on the cone $z^2 = x^2 + y^2$ that are closest to the point $(4, 2, 0)$.

6. Use Lagrange multipliers to find the maximum and minimum values of the function $f(x,y)=x^2+y$ subject to the constraint $x^2+y^2=4$.

7. Suppose a company makes x units of item A and y units of item B. Suppose also that their profit is given by the function $P(x, y) = 7x^2y - 520x + 2000y$. If the company must make exactly 39 total units how many units of item A and B should it make to maximize its profit?

8. Suppose you live on the surface $z^2 = x^2 + y^2$. The surface is heated by energy from the sun and the temperature at any given point is given by $T(x, y, z) = x^2 y - z$. Use the Lagrange multiplier method to find the location on the surface with the lowest temperature.